

Program-10: Kubernetes Dashboard and CLI for Cluster Monitoring.

Prerequisites :

- Docker
- Minikube
- kubectl

Step 1: Create Project Folder

```
mkdir Program-10
```

```
cd Program-10
```

step2:Start Minikube

```
minikube start --cpus=2 --memory=4096
```

step3:Enable Kubernetes Dashboard

```
minikube dashboard
```

step4:nano deployment.yaml

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-deployment
```

```
  labels:
```

```
    app: nginx
```

```
spec:
```

```
  replicas: 2
```

```
  selector:
```

```
    matchLabels:
```

```
      app: nginx
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: nginx
```

```
    spec:
```

```
      containers:
```

```
        - name: nginx
```

```
          image: nginx:latest
```

```
          ports:
```

```
        - containerPort: 80
```

step5:nano service.yaml

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: nginx-service
```

```
spec:
```

```
  selector:
```

```
    app: nginx
```

```
  ports:
```

```
    - protocol: TCP
```

port: 80
targetPort: 80
type: NodePort

Step 6: Apply Resources

```
kubectl apply -f deployment.yaml  
kubectl apply -f service.yaml
```

step7:Check Pods,nodes,svc

```
kubectl get pods  
kubectl get nodes  
kubectl get svc
```

Step 8: Check Pod Logs

```
kubectl get pods  
kubectl logs <pod-name>
```

Step 9: Describe Pod

```
kubectl describe pod <pod-name>
```

Step 10: Access Application in Browser

```
minikube service nginx-service
```

step11:Verify in Dashboard

Minikube Dashboard

